Book of Proof: Condensed Version with All Exercise Solutions

KS

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Part I
Fundamentals

1 Introduction to Sets

All of mathematics can be described with sets. Set theory is *a* perfectly suitable language to express and describe all types of mathematical structures.

Think of a *set* a collection of things. The things are called *elements* of the set. It's somewhat worth noting here that we're only concerned with sets that express mathematical entities like functions, numbers, points, etc. Set construction: $\{2, 3, 4, 5\}$ is a set which has 4 elements. Some sets have infinitely many elements, *not all* sets are finite.

We say that two sets are *equal* if they contain exactly the same elements. So: $\{2, 3, 4, 5\} = \{2, 4, 3, 5\}$ but $\{2, 3, 4, 5\} \neq \{2, 4, 6, 8\}$.

To express that an element is *member* of a set, we write $2 \in A$, where A is a set containing 2 as an element. Some sets are very important and we reserve special symbols for them, like:

$$\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{Q}, \mathbb{C}$$

Sets don't have to contain only numbers, for example $B = \{T, F\}$ or $C = \{a, b, c, d, e\}$, or $D = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$. A set can have *other sets as elements*:

$$K = \{1, \{2, 3\}, \{2, 4\}\}$$

It's really important to note here that K has 3 elements. $1, \{2, 3\}, \{2, 4\} \in K$ but $2, 3, 4 \notin K$.

If a set X is finite, we say that X's *cardinality* is the number of elements it has, and this number is denoted by $|X|$. An example of a special set is $\varnothing$. It is called the empty set and it enjoys the unique status of having no members and thus $|\varnothing| = 0$.

We have special *set builder notation* to describe sets that are too large or too complex to list between braces. Consider the infinite set of even integers $E = \{\ldots, -4, -2, 2, 4, 6 \ldots\}$. We can instead write

$$E = \{2n | n \in \mathbb{Z}\}.$$

We read it "the set of all things of form" $2n$, | "such that", n "is a member of" the set of integers. The following generalised structure follows for set-builder notation:

$$X = \{\text{expression} : \text{rule}\};$$

where the elements of X are understood to be *all values of expression* that are specified by the *rule*.

Example 1.0.1.

1. $\{n : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17...\}$
2. $\{n \in \mathbb{N} : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17\}$
3. $\{x \in \mathbb{R} : x^2 - 2 = 0\} = \{\sqrt{2}, \sqrt{-2}\}$
4. $\{x \in \mathbb{Z} : x^2 - 2 = 0\} = \varnothing$
5. $\{x \in \mathbb{Z} : |2x| < 4\} = \{-1, 0, 1\}$

Example 1.0.2. Describe the set $A = \{7a + 3b : a, b \in \mathbb{Z}\}$.

Solution. The set A is equivalent to the expression $7a + 3b$, where the rule for finding a, b is that they are members of the set of integers. Consider which integers. Let n be any integer, then $n = 7n + 3(-2n)$, such that $n = a$ and $b = -2n$. So, $n \in A$. It follows that A contains only integers, and furthermore, *every integer is an element of* A. So $A = \mathbb{Z}$. □

Important: Set-builder notation for $\mathbb{Q}$:

$$\mathbb{Q} = \{x : x = \frac{m}{n} \text{ where } m, n \in \mathbb{Z} \wedge n \neq 0\}$$

1.1 Exercises from Section 1

Exercise 1.1. Write each of the following sets by listing their elements between braces.

$1.\{5x - 1 : x \in \mathbb{Z}\}$ $9.\{x \in \mathbb{R} : \sin \pi x = 0\}$
$2.\{3x + 2 : x \in \mathbb{Z}\}$ $10.\{x \in \mathbb{R} : \cos x = 1\}$
$3.\{x \in \mathbb{Z} : -2 \leq x \leq 7\}$ $11.\{x \in \mathbb{Z} : |x| < 5\}$
$4.\{x \in \mathbb{N} : -2 < x \leq 7\}$ $12.\{x \in \mathbb{Z} : |2x| < 5\}$
$5.\{x \in \mathbb{R} : x^2 = 3\}$ $13.\{x \in \mathbb{Z} : |6x| < 5\}$
$6.\{x \in \mathbb{R} : x^2 = 9\}$ $14.\{5x : x \in \mathbb{Z}, |2x| \leq 8\}$
$7.\{x \in \mathbb{R} : x^2 + 5x = -6\}$ $15.\{5a + 2b : a, b \in \mathbb{Z}\}$
$8.\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\}$ $16.\{6a + 2b : a, b \in \mathbb{Z}\}$.

[Go to Solution]

Exercise 1.2. Write each of the following sets in set-builder notation.

$17.\{2, 4, 8, 16, 32, 64, \dots\}$ $23.\{3, 4, 5, 6, 7, 8\}$
$18.\{0, 4, 16, 36, 64, 100, \dots\}$ $24.\{-4, -3, -2, -1, 0, 1, 2\}$
$19.\{\dots, -6, -3, 0, 3, 6, 9, 12, 15\}$ $25.\{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \dots\}$
$20.\{\dots, -8, -3, 2, 7, 12, 17, \dots\}$ $26.\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, \dots\}$
$21.\{0, 1, 4, 9, 16, 25, 36, \dots\}$ $27.\{\dots, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}, \dots\}$
$22.\{3, 6, 11, 18, 27, 38, \dots\}$ $28.\{\dots, -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, \dots\}$.

[Go to Solution]

Exercise 1.3. Find the following cardinalities.

$29.|\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}|$ $34.|\{x \in \mathbb{N} : |x| < 10\}|$
$30.|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}|$ $35.|\{x \in \mathbb{Z} : x^2 < 10\}|$
$31.|\{\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}\}|$ $36.|\{x \in \mathbb{N} : x^2 < 10\}|$
$32.|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}\}|$ $37.|\{x \in \mathbb{N} : x^2 < 0\}|$
$33.|\{x \in \mathbb{Z} : |x| < 10\}|$ $38.|\{x \in \mathbb{N} : 5x \leq 20\}|$.

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Exercise 1.4. Sketch the following sets of points in the xy-plane.

$$39.\{(x,y) : x \in [1,2], y \in [1,2]\} \qquad 46.\{(x,y) : x, y \in \mathbb{R}, x^2 + y^2 \leq 1\}$$
$$40.\{(x,y) : x \in [0,1], y \in [1,2]\} \qquad 47.\{(x,y) : x, y \in \mathbb{R}, y \geq x^2 - 1\}$$
$$41.\{(x,y) : x \in [-1,1], y = 1\} \qquad 48.\{(x,y) : x, y \in \mathbb{R}, x > 1\}$$
$$42.\{(x,y) : x = 2, y \in [0,1]\} \qquad 49.\{(x, x+y) : x \in \mathbb{R}, y \in \mathbb{Z}\}$$
$$43.\{(x,y) : |x| = 2, y \in [0,1]\} \qquad 50.\{(x, \frac{x^2}{y}) : x \in \mathbb{R}, y \in \mathbb{N}\}$$
$$44.\{(x,x^2) : x \in \mathbb{R}\} \qquad 51.\{(x,y) \in \mathbb{R}^2 : (y-x)(y+x) = 0\}$$
$$45.\{(x,y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\} \qquad 52.\{(x,y) \in \mathbb{R}^2 : (y-x^2)(y+x^2) = 0\}.$$

[Go to Solution]

2 The Cartesian Product

Think of the Cartesian product as a type of multiplication.

$$A \times B$$

In order to understand products better we will define an *ordered pair*.

Definition 2.0.1. An ordered pair is a list of the form x, y of two things, namely x and y, enclosed in parenthesis and separated by a comma.[1]

There are many ordered pairs, consider the ordered pair $(2,4)$. Now, if we swap the numbers around, that would be an ordered pair as well, i.e. $(4,2)$. However, these two pairs are not identical, meaning

$$(2,4) \neq (4,2).$$

Ordered pairs can be used to describe points on a plane (as it is done in Calculus and Linear Algebra for example). Similarly to our previous talk about sets, ordered pairs need not be *just* numbers, they can be letters as well.

Example 2.0.1. The set $(\{2,3\},\{3,4\})$ can be an ordered pair of two sets, say A, B, then $(\{2,3\},\{3,4\}) = (A,B)$;
α, β can also be a pair; finally, $((2,3),(3,4))$ can be an ordered pair of an ordered pair.

So, the following definition of Cartesian Products is important:

Definition 2.0.2. The Cartesian Product of two sets A, B is a set, denoted by

$$A \times B$$

and defined as

$$A \times B = \{(a,b) : a \in A, b \in B\}.$$

[1] Additionally, one can use the Kuratowski ordered pair definition, i.e. $\{\{x\},\{x,y\}\} = (x,y)$. However, the one given above, even though slightly cryptic is easier to grasp for someone who has never encountered sets before.

The following is a more visual example of how ordered pairs are formed.

Example 2.0.2. Let $A = \{k, l, m\}, B = \{o, p\}$ for A, B some sets, then the product of A with B is

$$A \times B = \{(k, o), (k, p), (l, o), (l, p), (m, o), (m, p)\}.$$

Schematically, one can think of these pairs in terms of rows and columns or in terms of a corrdinate system where A is the x-axis and B is the y-axis.

The following is a fact:

Fact 2.0.1. If A, B are finite sets, then $|A \times B| = |A| \cdot |B|$.

Example 2.0.3. Let $A = \{1, 2, 3, 4, 5, 6\}$ and suppose these are dice values. The Cartesian Product of A with itself is

$$A \times A = A^2.$$

The product of A with itself will be a finite set of ordered pairs.

$$|A \times A| = |A| \cdot |A| = 6 \cdot 6 = 36.$$

Think of the product as rolling a dice twice an recording the outcome, where the first roll corresponds to the first result and the second to the second result.

Example 2.0.4. Another few very useful examples are the sets:

$$\begin{aligned} \mathbb{R} \times \mathbb{R} &= \{(x, y) : x, y \in \mathbb{R}\} \\ \mathbb{R} \times \mathbb{N} &= \{(x, y) : x \in \mathbb{R}, y \in \mathbb{N}\} \\ \mathbb{N}^2 &= \{(x, y) : x, y \in \mathbb{N}\}. \end{aligned}$$

The first example is best described as a coordinate system or an xy-plane. The second example is simply all points on the plane, whose second coordinate is a natural number. The third example, is best described as the first quadrant of the xy-plane, where every x, y are natural numbers.

It is possible for one factor of a Cartesian product to be a Cartesian Product itself. Suppose the following:

$$\mathbb{R} \times (\mathbb{N} \times \mathbb{Z}) = \{x, (y, z) : x \in \mathbb{R}, (y, z) \in \mathbb{N} \times \mathbb{Z}\}.$$

There is a different way to think about these pairs. They are called *ordered triples*. The brackets are optional in the above example, one can also write

$$\mathbb{R} \times \mathbb{N} \times Z = \{(x, y, z) : x \in \mathbb{R}, y \in \mathbb{N}, z \in Z\}.$$

In general we can define

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, x_2, \ldots, x_n) : x_1 \in A_1, x_2 \in A_2, \ldots, x_n \in A_n\},$$

or more succinctly,

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, x_2, \ldots, x_n) : x_i \in A_i\} \text{ for } i = \{1, 2, 3, \ldots, n\}$$

Example 2.0.5. Represent the two sides of a coin by the set $S = (H, T)$, the possible outcomes of tossing the coin seven times in a row can be described as:

Solution. This is the representation

$$S^7 = \{(x_1, x_2, \ldots, x_7) : x_1, x_2, \ldots, x_7 \in S\}.$$

Now to find the possible outcomes:

$$|S^7| = 7 \cdot 7 \cdot \ldots 7 = 128.$$

□

I believe the following to be clear but still worth noting. Notice that the cardinality of the product of S with itself seven times is written inside the || and not outside. This might seem like a silly thing to say, however syntax is very important.

2.1 Exercises from Section 2

Exercise 2.1. Write out the indicated sets by listing their elements between braces.

1. Suppose $A = \{1, 2, 3, 4\}, B = \{a, c\}$.

   $(a) A \times B$
   $(b) B \times A$
   $(c) A \times A$
   $(d) B \times B$
   $(e) \varnothing \times B$
   $(f) (A \times B) \times B$
   $(g) A \times (B \times B)$
   $(h) B^3$

2. Suppose $A = \{\pi, e, 0\}, B = \{0, 1\}$.

   $(a) A \times B$
   $(b) B \times A$
   $(c) A^2$
   $(d) B^2$
   $(e) A \times \varnothing$
   $(f) (A \times B) \times B$
   $(g) A \times (B \times B)$
   $(h) A \times B \times B.$

3. $\{x \in \mathbb{R} : x^2 = 2\} \times \{a, c, e\}$.

4. $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\}$.
5. $\{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$.
6. $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\}$.
7. $\{\varnothing\} \times \{0, \varnothing\} \times \{0, 1\}$.
8. $\{0, 1\}^4$.

[Go to Solution]

Exercise 2.2. Sketch these Cartesian Products on the xy-plane (or in $\mathbb{R}^3$ for the last two).

$(9)\{1, 2, 3\} \times \{-1, 0, 1\}$	$(15)\{1\} \times [0, 1]$
$(10)\{-1, 0, 1\} \times \{1, 2, 3\}$	$(16)[0, 1] \times \{1\}$
$(11)[0, 1] \times [0, 1]$	$(17)\mathbb{N} \times \mathbb{Z}$
$(12)[-1, 1] \times [1, 2]$	$(18)\mathbb{Z}^2$
$(13)\{1, 1.5, 2\} \times [1, 2]$	$(19)[0, 1]^3$
$(14)[1, 2] \times \{1, 1.5, 2\}$	$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \times [0, 1]$

[Go to Solution]

3 Subsets

It is common that every element, or some elements of a set A are also elements of another set B. Consider the following:

$$A = \{1, 2, 3\}, B = \{0, 1, 2, 3, 4, 5\}.$$

In cases as this one, we say that $A \subseteq B$, we might even say that $A \subset B$ (since all elements of A are contained within B).

Definition 3.0.1. Suppose A, B are sets. If every element of A is also an element of B, then we say that $A \subseteq B$. We write $A \nsubseteq B$ if A is *not* a subset of B i.e., if it is not true that *every* element of A is also an element of B. So $A \nsubseteq B$ tells us that *at least one* element of the set on the left is not contained in the set on the right.

Example 3.0.1. Some key examples are the following:

1. $\{2, 3, 7\} \subseteq \{2, 3, 4, 5, 6, 7\}$;
2. $\{2, 3, 7\} \nsubseteq \{2, 3, 4, 5, 6\}$;
3. $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$;
4. $\mathbb{R} \times \mathbb{N} \subseteq \mathbb{R}^2$;
5. $A \subseteq A$, any set A is a subset of itself;
6. $\varnothing \subseteq \varnothing$.

These examples bring out an important fact about the empty set.

$$\boxed{\text{If } B \text{ is any set whatsoever, then } \varnothing \subseteq B}.$$

Fact 3.0.1. The empty set is a subset of all sets, that is, $\varnothing \subseteq B$ for any set B.

Proof. Let B be any set, then consider the definition of subset. Then if $\varnothing \not\subseteq B$ this would mean that there are *some* members of $\varnothing$ which are not members of B. However, we know that by definition $x \notin \varnothing$, or more formally

$$\varnothing = \{x : x \neq x\}.$$

Therefore, $\varnothing \subseteq B$. □

Comment: This proof technique is called *Proof by Contradiction* and it does come up a lot in the exercises later on, so it is worth trying to understand what is happening at this stage, however, the proof strategy will be explained in detail later.

Fact 3.0.2. If $|B| = n$, then B has 2^n subsets. Further, for *any finite* set B that has n number of elements, then B has 2^n subsets.

Example 3.0.2. Consider the subsets of the set $B = \{1, 2, \{1, 3\}\}$, there are 3 elements so 2^3.

$$\{\varnothing, \{1\}, \{2\}, \{\{1, 3\}\}, \{1, 2\}, \{1, 2, \{1, 3\}\}, \{1, \{1, 3\}\}, \{2, \{1, 3\}\}\}.$$

Important: It is really important to notice that $\{1, 3\}$ is **not** a subset of B. As in:

$$\{1, 3\} \not\subseteq B \text{ but } \{1, 3\} \in B, \text{ however, } \{\{1, 3\}\} \subseteq B.$$

3.1 Exercises from Section 3

Exercise 3.1. List all the subsets of the following sets:

1. $\{1, 2, 3, 4\}$
2. $\{1, 2, \varnothing\}$
3. $\{\{\mathbb{R}\}\}$
4. $\varnothing$
5. $\{\varnothing\}$
6. $\{\mathbb{R}, \mathbb{N}, \mathbb{Q}\}$
7. $\{\mathbb{R}, \{\mathbb{N}, \mathbb{Q}\}\}$
8. $\{\{0, 1\}, \{0, 1, \{2\}\}, \{0\}\}$

[Go to Solution]

Exercise 3.2. Write out the following sets by listing their elements between braces.

1. $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 2\}$

2. $\{X \subseteq \mathbb{N} : |X| \leq 1\}$
3. $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 4\}$
4. $\{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 1\}$

[Go to Solution]

Exercise 3.3. Decide if the following statements are true or false. Explain.

1. $\mathbb{R}^3 \subseteq \mathbb{R}^3$
2. $\mathbb{R}^2 \subseteq \mathbb{R}^3$
3. $\{(x, y) \in \mathbb{R}^2 : x - 1 = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\}$
4. $\{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x - 1 = 0\}$

[Go to Solution]

4 Powersets

Definition 4.0.1. If A is a set, the *power set* of A is another set, denoted $\mathcal{P}(A)$ and defined to be the set of *all* subsets of A. Formally,

$$\mathcal{P}(A) = \{X : X \subseteq A\}.$$

Suppose that we have some set $A = \{1, 2, 3\}$, then

$$\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

A *finite* set of n-elements has w^n subsets and the powerset of that set has 2^n-elements.

Fact 4.0.1. If A is a finite set, then $|\mathcal{P}(A)| = 2^{|A|}$.

Example 4.0.1.

$$\mathcal{P}(\{a\}) \times \mathcal{P}(\{\varnothing\}) = \{(\{a\}, \{\varnothing\}), (\{a\}, \varnothing), (\varnothing, \varnothing), (\varnothing, \{\varnothing\})\}.$$

This example might appear a little bit intimidating, however consider the following:

$$\mathcal{P}(\{a\}) = \{\varnothing, \{a\}\} \quad \mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}.$$

Then we simply pair the elements together.

Next, some *wrong* facts about powersets.

1. $\mathcal{P}(1) = \{\varnothing, \{1\}\}$ is meaningless, since 1 is not a set. [2]
2. $\mathcal{P}(\{1, \{1, 2\}\}) = \{\varnothing, \{1\}, \{\{1, 2\}\}, \{1, \{1, 2\}\}\}$ becuase $\{1, 2\} \not\subseteq \{1, \{1, 2\}\}$, however $\{1, 2\} \in \mathcal{P}\{1, \{1, 2\}\}$.[3]
3. $\mathcal{P}(\{1, \{1, 2\}\})$ but $\{\{1\}\} \not\subseteq \{1, \{1, 2\}\}$, the reasoing is similar as in the previous footnote.

It is not possible to list all the members of a powerset of A if A is *infinite.*

[2] There could be some debate here, this is what is says in the textbook, so I will follow it, however we can express $1 = \{\{\varnothing\}\}$ or $\{\varnothing, \{\varnothing\}\}$ when we consider the V-hierarchy of set construction.

[3] The correct statement would be $\{\{1, 2\}\} \subseteq \mathcal{P}(\{1, \{1, 2\}\})$.

4.1 Exercises From Section 4

Exercise 4.1. Write the following sets by listing their elements between braces.

1. $\mathcal{P}(\{\{a,b\},\{c\}\})$
2. $\mathcal{P}(\{1,2,3,4\})$
3. $\mathcal{P}(\{\{\varnothing\},5\})$
4. $\mathcal{P}(\{\mathbb{R},\mathbb{Q}\})$
5. $\mathcal{P}(\mathcal{P}(\{2\}))$
6. $\mathcal{P}(\{1,2\})\times\mathcal{P}(\{3\})$
7. $\mathcal{P}(\{a,b\})\times\mathcal{P}(\{0,1\})$
8. $\mathcal{P}(\{1,2\}\times\{3\})$
9. $\mathcal{P}(\{a,b\}\times\{0\})$
10. $\{X\in\mathcal{P}(\{1,2,3\}):|X|\leq 1\}$
11. $\{X\subseteq\mathcal{P}(\{1,2,3\}):|X|\leq 1\}$
12. $\{X\in\mathcal{P}(\{1,2,3\}):2\in X\}$.

[Go to Solution]

Exercise 4.2. Suppose that $|A|=m,|B|=n$. Find the following cardinalities.

1. $|\mathcal{P}\mathcal{P}\mathcal{P}(A)|$
2. $|\mathcal{P}\mathcal{P}(A)|$
3. $|\mathcal{P}(A\times B)|$
4. $|\mathcal{P}(A)\times\mathcal{P}(B)|$
5. $|\{X\in\mathcal{P}(A):|X|\leq 1\}|$
6. $|\mathcal{P}(A\times\mathcal{P}(B))|$
7. $|\mathcal{P}\mathcal{P}\mathcal{P}(A\times\varnothing)|$
8. $|\{X\subseteq\mathcal{P}(A):|X|\leq 1\}|$.

[Go to Solution]

5 Union, Intersection, Difference

The same way numbers can be combined with operations such as addition, multiplication, subtraction, and division, we have similar operations for sets too.

Definition 5.0.1. Let A, B be sets then:

1. The *union* of A, B is $A \cup B = \{x : x \in A \vee x \in B\}$
2. The *intersection* of A, B is $A \cap B = \{x : x \in A \wedge x \in B\}$
3. The *difference* of A, B is $A - B = \{x : x \in A \wedge x \notin B\}$

Let us consider the following examples:

Example 5.0.1. Let $A = \{a, b, c, d, e\}, B = \{d, e, f\}, C = \{1, 2, 3\}$:

1. $A \cup B = \{a, b, c, d, e, f\}$
2. $A \cap B = \{d, e, f\}$
3. $A - B = \{a, b, c\}$
4. $B - A = \{f\}$
5. $(A - B) \cup (B - A) = \{a, b, c\} \cup \{f\} = \{a, b, c, f\}$
6. $A \cup C = \{a, b, c, d, e, 1, 2, 3\}$
7. $A \cap C = \varnothing$
8. $A - C = \{a, b, c, d, e\}$
9. $(A \cap C) \cup (A - C) = \varnothing \cup \{a, b, c, d, e\} = \{a, b, c, d, e\}$
10. $(A \cap B) \times B = \{d, e\} \times \{d, e, f\} = \{(d, d), (d, e), (d, f), (e, d), (e, e), (e, f)\}$
11. $(A \times C) \cap (B \times C) = \{(d, 1), (d, 2), (d, 3), (e, 1), (e, 2), (e, 3)\}$.

Important: It is always true that $X \cap Y = Y \cap X, X \cup Y = Y \cup X$. In other words, these are *commutable operations.* However, in general, $X - Y \neq Y - X$.

5.1 Exercises and Solutions for Section 5

Part II
Answers to All Exercises

6 Solutions to Exercises From Section 1

Solution. 1.1

1. $\{5x-1 : x \in \mathbb{Z}\} = \{\dots, -6, -1, 4, 9, \dots\}$
2. $\{3x+2 : x \in \mathbb{Z}\} = \{\dots, -1, 2, 5, 8\}$.
3. $\{x \in \mathbb{Z} : -2 \leq x \leq 7\} = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$.
4. $\{x \in \mathbb{N} : -2 < x \leq 7\} = \{0, 1, 2, 3, 4, 5, 6, 7\}$.
5. $\{x \in \mathbb{R} : x^2 = 3\} = \{-\sqrt{3}, \sqrt{3}\}$.
6. $\{x \in \mathbb{R} : x^2 = 9\} = \{-3, 3\}$.
7. $\{x \in \mathbb{R} : x^2 + 5x = -6\} = \{-2, -3\}$.
8. $\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\} = \{-3, -2, 0\}$.
9. $\{x \in \mathbb{R} : \sin \pi x = 0\} = \mathbb{Z}$.
10. $\{x \in \mathbb{R} : \cos x = 1\} = \{2n\pi : n \in \mathbb{Z}\}$.
11. $\{x \in \mathbb{Z} : |x| < 5\} = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$.
12. $\{x \in \mathbb{Z} : |2x| < 5\} = \{-2, -1, 0, 1, 2\}$.
13. $\{x \in \mathbb{Z} : |6x| < 5\} = \{0\}$.
14. $\{5x : x \in \mathbb{Z}, |2x| \leq 8\} = \{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$.
15. $\{5a + 2b : a, b \in \mathbb{Z}\} = \mathbb{Z}$.
16. $\{6a + 2b : a, b \in \mathbb{Z}\} = 2\mathbb{Z}$.

□

Solution. 1.2 *Tip:* Some problems might look intimidating but in general, the easiest way to complete a problem like this, would be to consider the following three steps.

1 Is there common denominator?

2 Is there a constant that can be factored out?

3 What remains and how can this be generalised?

With those tips in mind, we proceed to the solutions.

17 $\{2, 4, 8, 16, 32, 64\} = \{2^x : x \in \mathbb{N}\}$

18 $\{0, 4, 16, 36, 64, 100, \dots\} = \{x^2 : x \in \mathbb{N}, 2x\}$.

19 $\{\dots, -6, -3, 0, 3, 6, 9, 12, 15, \dots\} = \{3x : x \in \mathbb{Z}\}$.

20 $\{\dots, -8, -3, 2, 7, 12, 17, \dots\} = \{5x : x \in \mathbb{Z}\}$.

21 $\{0, 1, 4, 9, 16, 25, 36, \dots\} = \{2x : x \in \mathbb{N}\}$

22 $\{3, 6, 11, 18, 27, 38, \dots\} = \{x^2 + 2 : x \in \mathbb{Z}^+\}$

23 $\{3, 4, 5, 6, 7, 8\} = \{3 \leq x \leq 8 : x \in \mathbb{N}\}$

24 $\{-4, -3, -2, -1, 0, 1, 2\} = \{-4 \leq x \leq 2 : x \in \mathbb{Z}\}$

25 $\{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \dots\} = \{2^{-x} : x \in \mathbb{Z}\}$

26 $\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27\} = \{3^{-x} : x \in \mathbb{Z}\}$

27 $\{\dots, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}\} = \{\frac{n\pi}{2} : n \in \mathbb{Z}\}$

28 $\{\dots, -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, \dots\} = \{\frac{3x}{4} : x \in \mathbb{Z}\}$

□

Solution. 1.3

29 $|\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}| = |3|$

30 $|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}| = |5|$

31 $|\{\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}\}| = |2|$

32 $|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}\}| = |1|$

33 $|\{x \in \mathbb{Z} : |x| < 10\}| = |19|$

34 $|x \in \mathbb{N} : |x| < 10| = |10|$

35 $|\{x \in \mathbb{Z} : x^2 < 10\}| = |8|$

36 $|\{x \in \mathbb{N} : x^2 < 10\}| = |4|$

37 $|\{x \in \mathbb{N} : x^2 < 10\}| = \varnothing$

38 $|\{x \in \mathbb{N} : 5x \leq 20\}| = |5|$.

□

Solution. 1.4

39 $\{(x, y) : x \in [1, 2], y \in [1, 2]\} \Rightarrow$ This would be a solid 1×1 square in $\mathbb{R}^2$ with corners $(1, 1), (1, 2), (2, 1), (2, 2)$

40 $\{(x, y) : x \in [0, 1], y \in [1, 2]\} \Rightarrow$ This would be a solid 1×1 square in $\mathbb{R}^2$ with corners $(0, 1), (0, 2), (1, 1), (1, 2)$.

41 $\{(x, y) : x \in [-1, 1], y = 1\} \Rightarrow$ This would be a line connecting $(-1, 1)$ with $(1, 1)$.

42 $\{(x, y) : x = 2, y \in [0, 1]\} \Rightarrow$ This would be a line connecting $(2, 0), (2, 1)$

43 $\{(x, y) : |x| = 2, y \in [0, 1]\} \Rightarrow$ This would be two vertical line segments, that connect $(-2, 0) - (2, 1)$ and the other one would be $(2, 0) - (2, 1)$.

44 $\{(x, x^2) : x \in \mathbb{R}\} \Rightarrow$ This would be the parabolla $y = x^2$.

45 $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\} \Rightarrow$ This would be the unit circle centered at the origin $(0, 0)$ with $r = 1$

46 $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 \leq 1\} \Rightarrow$ This would be a disk with $r = 1$, centered at $(0, 0)$.

47 $\{(x, y) : x, y \in \mathbb{R}, y \leq x^2 - 1\} \Rightarrow$ This would be the region underneath the parabola $y = x^2 - 1$.

48 $\{(x, y) : x, y \in \mathbb{R}x > 1\} \Rightarrow$ This would the whole upper right and down left quadrants of the xy-plane, or an *open half-plane*, starting from $x = 2$.

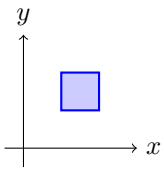
49 $\{(x, x + y) : x \in \mathbb{R}, y \in \mathbb{Z}\} \Rightarrow$ This would be infinitely many parallel lines with slope 1.

50 $\{(x, \frac{x^2}{y})x \in \mathbb{R}, y \in \mathbb{N}\} \Rightarrow$ Division by 0 is undefined, so we skip those values. Notice that the relation between the first and second coordinate is quadratic, instead of linear. That is to say $v = \frac{u^2}{k}$, so we have a upward-opening parabolas passing through the origin

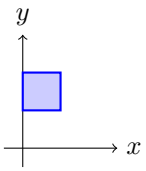
51 $\{(x, y) \in \mathbb{R}^2 : (y - x)(y + x) = 0\} \Rightarrow$ Here we have lines passing through the origin in an X shape.

52 $\{(x, y) \in \mathbb{R}^2 : (y - x^2)(y + x^2) = 0\} \Rightarrow$ Lastly, we have two parabolas touching at the origin one opening upwards, one downwards, $y = x^2, y = -x^2$, respectively.

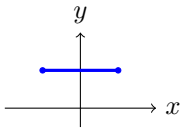
□



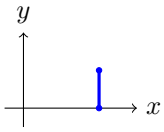
(a) Item 39



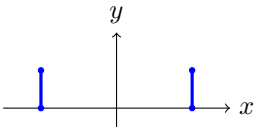
(b) Item 40



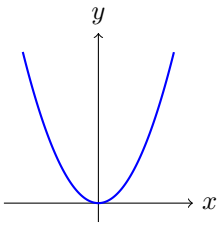
(c) Item 41



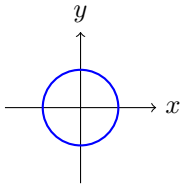
(d) Item 42



(e) Item 43

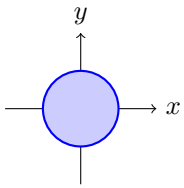


(f) Item 44

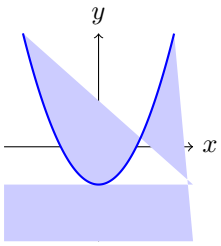


(g) Item 45

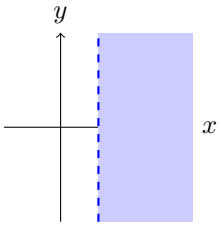
Figure 1: Geometric Representations 39–45



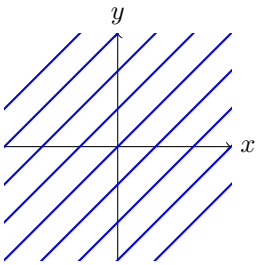
(h) Item 46



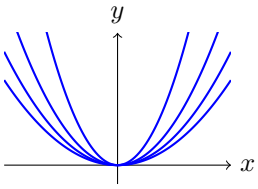
(i) Item 47



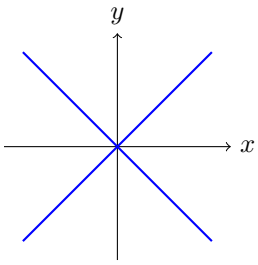
(j) Item 48



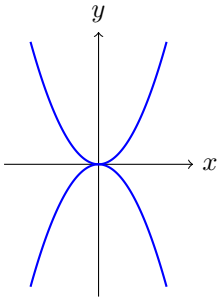
(k) Item 49



(l) Item 50



(m) Item 51



(n) Item 52

Figure 1: Geometric Representations 46–52 (Continued)

7 Solutions to Exercises from Section 2

Solution. 2.1

$(a) A \times B = \{(1,a),(1,c),(2,a),(2,c),(3,a),(3,c),(4,a),(4,c)\}$

$(b) B \times A = \{(a,1),(c,1),(a,2),(c,2),(a,3),(c,3),(a,4),(c,4)\}$

$(c) A \times A = A^2 = \{(1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),$

$(2,4),(3,1),(3,2),(3,3),(3,4),(4,1),(4,2),(4,3),(4,4)\}$
$(d) B \times B = B^2 = \{(a,a),(a,c)\}$

$(e) \varnothing \times B = \{(a,b) : a \in \varnothing, b \in B\} = \varnothing$

$(f)(A \times B) \times B = \{(1,a),(1,c),(2,a),(2,c),(3,a),(3,c),(4,a),(4,c)\} \times \{a,c\} =$
$\{((1,a),a),((1,c),a),((2,a),a),$
$((2,c),a),((3,a),a),((3,c),a),$
$((4,a),a),((4,c),a),((1,a),c),$
$((1,c),c),((2,a),c),((2,c),c),$
$((3,a),c),((3,c),c),((4,a),c),((4,c),c)\}$

$(g) A \times (B \times B) = \{1,2,3,4\} \times \{(a,a),(a,c)\} =$
$\{(1,(a,a)),(1,(a,c)),(2,(a,a)),(2,(a,c)),$
$(3,(a,a)),(3,(a,c)),(4,(a,a)),(4,(a,c))\}$
$(h) B^3 = \{(a,a,a),(a,a,c),(a,c,a),(a,c,c),(c,a,a),(c,a,c),(c,c,a),(c,c,c)\} = |2^3| = 8$

This ends Part I of 2.1. Now, for part II.

$(a) A \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\}$

$(b) B \times A = \{(0, \pi), (1, \pi), (0, e), (1, e), (0, 0), (1, 0)\}$

$(c) A^2 = \{(\pi, \pi), (\pi, e), (\pi, 0)$
$(e, \pi), (e, e), (e, 0)$
$(0, \pi), (0, e), (0, 0)\}$

$(d) B^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

$(e) A \times \varnothing = \varnothing$

$(f)(A \times B) \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\} \times \{0, 1\}$
$\{(\pi, (0, 0)), (\pi, (0, 1)), (\pi, (1, 0)),$
$(\pi, (1, 1,)), (e, (0, 0)), (e, (0, 1)),$
$(e, (1, 0)), (e, (1, 1)), (0, (0, 0)),$
$(0, (0, 1)), (0, (1, 0)), (0(1, 1))\}$

$(g) A \times (B \times B) = \{\pi, e, 0\} \times \{(0, 0), (0, 1), (1, 0), (1, 1)\}$
$\{(\pi, (0, 0)), (\pi, (0, 1)), (\pi, (1, 0)), (\pi, (1, 1)),$
$(e, (0, 0)), (e, (0, 1)), (e, (1, 0)), (e, (1, 1))$
$(0, (0, 0)), (0, (0, 1)), (0, (1, 0)), (0, (1, 1))\}$

$(h) A \times B \times B = \{(\pi, 0, 0), (\pi, 0, 1), (\pi, 1, 0),$
$(\pi, 1, 1), (e, 0, 0), (e, 0, 1),$
$(e, 1, 0), (e, 1, 1), (0, 0, 0),$
$(0, 0, 1), (0, 1, 0), (0, 1, 1)\}$

This concludes part II. Lastly, part III.

(3) $\{x \in \mathbb{R} : x^2 = 2\} \times \{a, c, e\} =$
$\{-\sqrt{2}, \sqrt{2}\} \times \{a, c, e\} =$
$\{(-\sqrt{2}, a), (-\sqrt{2}, c), (-\sqrt{2}, e),$
$(\sqrt{2}, a), (\sqrt{2}, c), (\sqrt{2}, e)\}$

(4) $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\} =$
$\{3, 4\} \times \{-5, 5\} = \{(3, -5), (3, 5), (4, -5), (4, 5)\}$

(5) $\{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\} =$
$\{-\sqrt{2}, \sqrt{2}\} \times \{-2, 2\} = \{(-\sqrt{2}, -2), (-\sqrt{2}, 2), (\sqrt{2}, -2), (\sqrt{2}, 2)\}$

(6) $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\} =$
$\{-1, 0, 1\} \times \{0, 1\} =$
$\{(-1, 0), (-1, 1), (0, 0), (0, 1), (1, 0), (1, 1)\}$

(7) $\{\varnothing\} \times \{0, \varnothing\} \times \{0, 1\} =$
$\{(\{\varnothing\}, 0, 0), (\{\varnothing\}, \varnothing, 0), (\{\varnothing\}, 0, 1), (\{\varnothing\}, \varnothing, 1)\}$

(8) $\{0, 1\}^4 = \{(0, 0, 0, 0), (0, 0, 0, 1), (0, 0, 1, 0), (0, 1, 0, 0),$
$(1, 0, 0, 0), (0, 0, 1, 1), (0, 1, 0, 1), (1, 0, 0, 1),$
$(0, 1, 1, 0), (1, 1, 0, 0), (1, 1, 1, 0), (1, 1, 1, 1),$
$(0, 1, 1, 1), (1, 0, 1, 0), (1, 0, 1, 1), (1, 1, 0, 1)\}.$

□

Solution. 2.2 As in the previous exercise where we sketched out the sets on the xy-plane, we will

first solve them/explain them and then go on to sketch them.

(9)$\{1,2,3\} \times \{-1,0,1\} =$
$\{(1,-1),(1,0),(1,1),$
$(2,-1),(2,0),(2,1),$
$(3,-1),(3,0),(3,1)\}$

(10)$\{-1,0,1\} \times \{1,2,3\} =$
$\{(-1,1),(-1,2),(-1,3),$
$(0,1),(0,2),(0,3),$
$(1,1),(1,2),(1,3)\}$

(11)$[0,1] \times [0,1]$ these would be all the paired points in that interval starting from
$x = 0, y = 1$

(12)$[-1,1] \times [1,2]$ these would be all the paired points in that interval starting from
$x = -1, y = 1$ and filling the space up to $x = 1, y = 2$

(13)$\{1, 1.5, 2\} \times [1,2]$ similarly to the cases above, those will become more obvious when we sketch them out

(14)$[1,2] \times \{1, 1.5, 2\},$ ditto.

(15)$\{1\} \times [0,1]$

(16)$[0,1] \times \{1\}$

(17)$\mathbb{N} \times \mathbb{Z},$ here, we take the natural numbers as our first coordinate and the integers as our second.

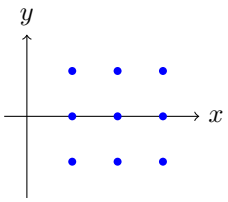
(18)$\mathbb{Z}^2,$ here we have the whole xy-plane with whole positive and negative integers.

(19)$[0,1]^2$

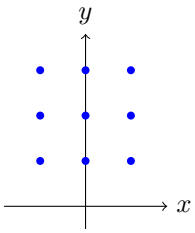
(20)$\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \times [0,1].$

Now, we will show them plotted on the xy-plane.

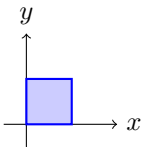
□



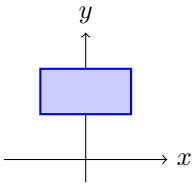
(a) Item (9)



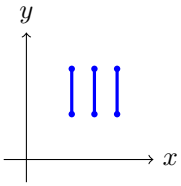
(b) Item (10)



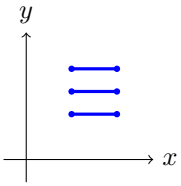
(c) Item (11)



(d) Item (12)

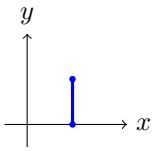


(e) Item (13)

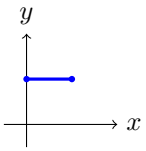


(f) Item (14)

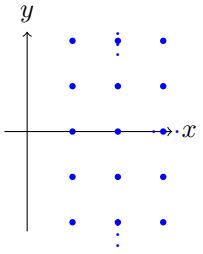
Figure 2: Plots for Cartesian Products (Items 9–14)



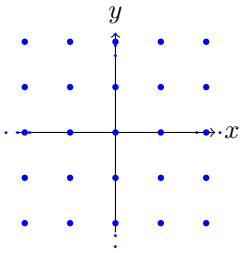
(g) Item (15)



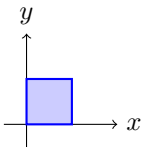
(h) Item (16)



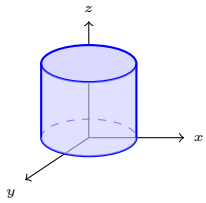
(i) Item (17) $\mathbb{N} \times \mathbb{Z}$



(j) Item (18) $\mathbb{Z}^2$



(k) Item (19) $[0,1]^2$



(l) Item (20) Disk $\times [0,1]$

Figure 2: Plots for Cartesian Products (Items 15–20, Continued)

8 Solutions to Exercises from Section 3

Solution. 3.1

(1) $\{1,2,3,4\} =$
$\varnothing, \{1\}, \{2\}, \{3\}, \{4\},$
$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\},$
$\{2,4\}, \{3,4\}, \{1,2,3\}, \{1,2,4\},$
$\{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}$

(2) $\{1,2,\varnothing\} =$
$\varnothing, \{1\}, \{2\},$
$\{\varnothing\}, \{1,2\}, \{1,\varnothing\},$
$\{2,\varnothing\}, \{1,2,\varnothing\}$

(3) $\{\{\mathbb{R}\}\} = \varnothing, \{\mathbb{R}\}$

(4) $\varnothing = \varnothing$

(5) $\{\varnothing\} = \varnothing, \{\varnothing\}$

(6) $\{\mathbb{R}, \mathbb{Q}, \mathbb{N}\} =$
$\varnothing, \{\mathbb{R}\}, \{N\}, \{\mathbb{Q}\},$
$\{\mathbb{R}, \mathbb{N}\}, \{\mathbb{R}, \mathbb{Q}\}, \{\mathbb{N}, \mathbb{Q}\}, \{\mathbb{R}, \mathbb{N}, \mathbb{Q}\}$

(7) $\{\mathbb{R}, \{\mathbb{N}, \mathbb{Q}\}\} =$
$\varnothing, \{\mathbb{R}\}, \{\mathbb{N}, \mathbb{Q}\}, \{\mathbb{R}, \{\mathbb{N}, \mathbb{Q}\}\}$

(8) $\{\{0,1\}, \{0,1,\{2\}\}, \{0\}\} =$
$\varnothing, \{0,1\}, \{0,1,\{2\}\}, \{0\},$
$\{\{0,1\}, \{0,1,\{2\}\}\}, \{\{0,1\}, \{0\}\},$
$\{\{0,1,\{2\}\}, \{0\}\}, \{\{0,1\}, \{0,1,\{2\}\}, \{0\}\}.$

□

Solution. 3.2

$$(1)\ \{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 2\}$$

$$(2)\ \{X \subseteq \mathbb{N} : |X| \leq 1\}$$

$$(3)\ \{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 4\}$$

$$(4)\ \{X : X \subseteq \{3, 2, a\} \text{ and } |X| = 1\}$$

□

Solution. 3.3

$$(1)\ \mathbb{R}^3 \subseteq \mathbb{R}^3$$

(1) is true, for the fact that every set is a subset of itself.

$$(2)\ \mathbb{R}^2 \subseteq \mathbb{R}^3$$

(2) is tricky, strictly speaking it is false. However, if you have any experience with Linear Algebra you might consider it true. In Linear Algebra we abuse the notation and say that $\mathbb{R}^3$ is an extension of $\mathbb{R}^2$, addiing the third dimension to a two dimensional plane. In set theory however, we are a bit more strict. The above is false, since

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R} \wedge y \in \mathbb{R}\},$$

And

$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) : x \in \mathbb{R} \wedge y \in \mathbb{R} \wedge z \in \mathbb{R}\}.$$

So, the first *object* takes in pairs, and the second object takes in triples.

$$(3)\ \{(x, y) \in \mathbb{R}^2 : x - 1 = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\}.$$

Consider a small example here, since y is not present in the first set, we infer that $x = 1$, so $\{1\}$. In the second set we have $x^2 - x = 0$, solving this gives us $x = 0, \vee x = 1$, so $\{0, 1\}$. Then it follows that the example is true.

$$(4)\ \{(x, y) \in \mathbb{R}^2 : x^2 - x = 0\} \subseteq \{(x, y) \in \mathbb{R}^2 : x - 1 = 0\}.$$

This example is false, in the LHS, we can have $0^2 - 0 = 0$, whereas in the RHS we cannot, unless $x = 1$, but then $\{0, 0\} \not\subseteq \{1\}$. □

9 Solutions to Exercises from Section 4

Solution. 4.1

(1) $\mathcal{P}(\{\{a,b\},\{c\}\}) = \{\varnothing, \{\{a,b\}\}, \{\{c\}\}, \{a,b,\{c\}\}\} = 2^2 = 4$ elements

(2) $\mathcal{P}(\{1,2,3,4\}) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{4\},$
$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\},$
$\{3,4\}, \{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}\} = 2^4 = 16$ elements.

□

Solution. 4.2

□

10 Solutions to Exercises from Section 5

11 Solutions to Exercises from Section 6

12 Solutions to Exercises from Section 7

13 Solutions to Exercises from Section 8